

GEOG 5303

Geographic Information Science: Spatial Programming

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	4
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Machine Learning Engineer (73.7 / 100)	59.8 / 100

60

CURRENT AVG SCORE
across 6 job roles

92

POTENTIAL SCORE
with recommended additions

+40

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Focuses on the extension of geographic information systems (GIS) through programming as well as on the development of algorithms for spatial analysis and information extraction in vector and raster data using open source tools. Covers concepts, principles and techniques of programming and solving spatial problems in natural and social science settings. Group projects will foster skillsets in implementing solutions to complex spatial problems.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	73.7	100.0	+26.3	MEDIUM
AI Engineer	69.1	100.0	+30.9	HIGH
ML Engineer	60.6	83.2	+22.5	MEDIUM
Applied Scientist	56.3	96.3	+39.9	HIGH
Data Scientist	52.2	87.9	+35.7	HIGH
MLOps Engineer	46.9	83.7	+36.7	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Deep Learning 1 of 6 roles

training

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

2 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

73.7 → 100.0 (+26.3 pts)

web analytics exploratory business design models model applications

AI Engineer

69.1 → 100.0 (+30.9 pts)

perception models software model based technology design policy

ML Engineer

60.6 → 83.2 (+22.5 pts)

models model world building

Applied Scientist

56.3 → 96.3 (+39.9 pts)

biology computational research interdisciplinary models design real domains

Data Scientist

52.2 → 87.9 (+35.7 pts)

toolkit analytics different approaches

MLOps Engineer

46.9 → 83.7 (+36.7 pts)

model text training models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.